

Observatoire de Bretagne Sud

Rapid-response capabilities for transient alerts

Optical follow-up of GRB / GW / IceCube / EP-FXT / TNS alerts — on-demand imaging and differential photometry

Prepared for: MMA collaborators

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1. Facility at a glance

Site & infrastructure		Instrumentation	
Location	Southern Brittany (France, ~48°N, 3°W)	Telescope	Amateur-class reflector on GoTo mount
Altitude	Low-altitude coastal site	Detector	Cooled CMOS science camera
Typical seeing	2.0 – 3.5" FWHM	Filters	Gaia-like photometric set: G
Dark sky	Bortle 4 (rural)	BP	,
Operation mode	Fully robotic, remotely triggered	RP	,
		Guiding	Off-axis, closed-loop
		Plate solving	Astrometry.net (on-site, < 10 s)

Why this note matters. The system is built specifically to **trigger, acquire, reduce, plate-solve and publish photometry of a new transient within ~15 minutes** of an external alert, using a pipeline that matches the workflow of professional rapid-response networks (GOTO, MASTER, BOOTES) at a fraction of the cost — making it a complementary asset for multi-messenger follow-up in western Europe.

2. Alert-to-first-exposure timeline

End-to-end latency on a clear, already-open night. All steps are automated; no operator intervention is required between the alert ingestion and the first science-grade stacked image being pushed to STDWeb for photometric calibration.

Stage	Action	Typical time	Notes
T + 0 s	Alert received	—	Via GCN, AstroColibri, TNS or LVK/IceCube broker (JSON/VOEvent)
T + 10 s	Visibility & horizon check	< 10 s	Auto-compute altitude, moon separation, airmass; accept if alt > 25°
T + 30 s	Queue injection (ToO priority)	< 1 min	Current target interrupted; NINA sequencer receives the new block
T + 2 min	Slew & centre	60 – 120 s	GoTo slew + platesolve-and-recentre loop
T + 4 min	Autofocus (if > 1 h old)	≈ 2 min	Skipped when the last AF is recent
T + 6 min	Exposure sequence starts	—	Typically 10 × 120 s in G, BP, RP (or a single filter for deepest detection)

Stage	Action	Typical time	Notes
T + 12 min	First stacked <code>res.fit</code> written	≈ 6 min	Siril CLI: dark + flat + registration + mean stack (2×2 binned)
T + 15 min	Plate-solved + uploaded to STDWeb	≈ 3 min	Calibrated photometry, ZTF-DR7 reference, candidate list available to collaborators

Current best reaction time: ~15 minutes from alert to published photometric measurement, in **good night conditions** (dome already open, telescope tracking, recent autofocus, clear weather). Cold-start (dome opening, cool-down, focusing from scratch) adds ≈ **20 minutes**.

Known limiting factors. Weather, target below horizon, moon proximity < 20°, or manual safety hold. The facility does not operate in daylight and has no dome-opening robotic weather safety certified for unattended GW-scale tiling — ToOs are accepted from astronomical dusk to dawn on pre-scheduled clear nights.

3. Alert ingestion & target enrichment

The Night Manager exposes a REST endpoint (POST `/api/targets/too`) accepting a minimal JSON payload (name, ra, dec, priority, optional filters, exp_time, n_frames). Each accepted target is automatically enriched against:

- **TNS** — classification, redshift, host galaxy, discovery photometry
- **AstroColibri** — multi-messenger broker linking GW/GRB/v events via `/event?trigger_id=TNS<name>`
- **GCN notices** — parsed for EP-WXT/FXT, Swift-BAT, Fermi-GBM, LVK S-type, IceCube Gold/Bronze
- **Gaia DR3 / ZTF DR7** — cross-match for foreground contaminants and reference photometry

A dedicated crossmatch tool (`tools/grb_crossmatch.py`) takes the raw alert position + error circle (e.g. 20" for EP-FXT), queries Gaia DR3, runs Source-Extractor-like (`sep`) extraction on each new frame, and flags any source inside the error circle with **no Gaia counterpart** as a candidate optical afterglow — in near-real time, as frames land on NAS.

4. Detection limits vs. exposure

The numbers below are **empirical**, derived from 84 stacks acquired between 2026-03-19 and 2026-04-02 and photometrically calibrated against Gaia DR3 by STDWeb at **S/N = 5**. Each stack was normalised to a reference block of **10 × 120 s** (20 min total exposure) assuming a background-limited regime ($\Delta\text{mag} = 1.25 \times \log_{10}(N_{\text{ref}} / N_{\text{obs}})$).

$$\text{mag}_{\text{lim}}(\text{S/N}=5, N \times 120 \text{ s}) = \text{mag}_{\text{lim}}(10 \times 120 \text{ s}) + 1.25 \times \log_{10}(N / 10)$$

4.1 Observed S/N = 5 limits at 10 × 120 s (Gaia-equivalent filters)

Filter	n	Poor (10 th pct)	Typical (median)	Good (90 th pct)	Best observed
G	32	17.4	18.9	19.9	20.2
BP	27	18.4	19.0	20.2	20.2
RP	25	17.0	18.5	18.9	19.8

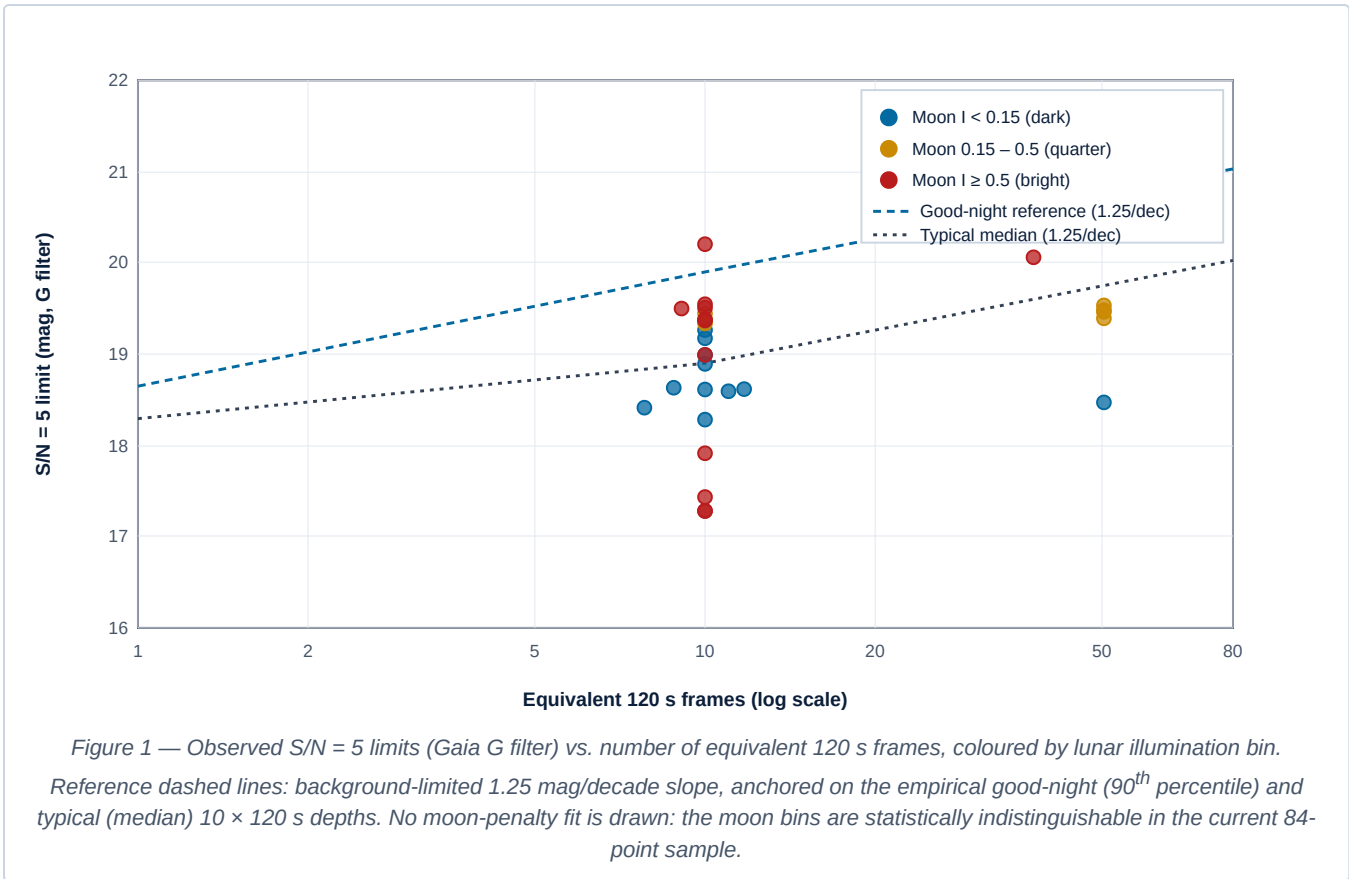
Poor nights are dominated by clouds, poor focus and bright-moon background. **Typical** is the realistic expectation for an operator-triggered ToO block on a clear night; **Good** requires stable transparency, good focus, and airmass < 1.5.

4.2 Scaling to other exposure counts

N × 120 s frames	Total exposure	G (typical)	G (good)	BP (good)	RP (good)
1	2 min	17.6	18.6	18.9	17.6
5	10 min	18.5	19.5	19.8	18.5
10	20 min	18.9	19.9	20.2	18.9
20	40 min	19.2	20.3	20.6	19.2
30	1 h	19.5	20.5	20.8	19.4
50	1.7 h	19.8	20.8	21.1	19.7
80	2.7 h	20.0	21.0	21.3	20.0
130	4.3 h	20.3	21.3	21.6	20.2
200	6.7 h	20.6	21.5	21.8	20.5

Scaling assumes the background-limited regime (1.25 mag per decade). Real gains saturate beyond ≈ 1 h for field transparency and systematic reasons (tracking drift, focus drift, sky gradients); use the numbers above as an upper bound.

Note on the lunar penalty. The current 84-point dataset is too small and too biased to measure the moon penalty — the deepest stacks in the series happen to have been taken around moon illumination $I \approx 0.7$, and the scatter from transparency / seeing / focus (~0.6 mag RMS) is larger than the expected moon signal. The medians per moon bin (G filter, 10 × 120 s) are: dark $I < 0.15 \rightarrow 18.65 \text{ mag}$ · quarter $\rightarrow 19.76 \text{ mag}$ · bright $I > 0.5 \rightarrow 19.54 \text{ mag}$, i.e. statistically indistinguishable. A robust moon-penalty coefficient will be published once a balanced, multi-lunation dataset is available. For now, assume a conservative 0.5 – 1.0 mag shallower limit within 30° of a > 50 % illuminated moon, consistent with typical rural-site Bortle-4 expectations.



4.3 Rules of thumb (clear night, airmass < 1.5)

- **Snapshot (1×120 s):** $G \approx 17.6$ typical, 18.6 on a good night.
- **Standard ToO block (10×120 s = 20 min):** $G \approx 18.9$ typical, **19.9** on a good night; BP ≈ 20.2 (good); RP ≈ 18.9 (good).
- **Deep stack (30×120 s = 1 h):** $G \approx 19.5$ typical, **20.5** on a good night.
- **Ultra-deep (80×120 s ≈ 2.7 h):** $G \approx 21.0$ on a good night (observed maximum so far: 20.7 mag).
- **Moon penalty:** not yet measurable; budget 0.5 – 1.0 mag shallower within 30° of a bright moon.

5. Science cases where the facility is relevant

Alert type	Typical mag	Feasibility / comment
GRB optical afterglow (early)	15 – 19	Easy at $T_0 + 15$ min; 1×120 s detects, 10×120 s gives colour.
GRB late afterglow ($T_0 +$ hours)	19 – 20.5	Feasible in deep mode on good nights ($30 - 80 \times 120$ s).
Kilonova (GW170817-like) @ 40 Mpc	17 – 19	Detection in first 2–3 days; light-curve sampling in G/BP/RP.
Kilonova @ 200 Mpc	21 – 22	Out of reach — below current best-observed depth (~ 20.7 mag) even on the best nights.
IceCube v (blazar flare follow-up)	16 – 19	Nominal; nightly monitoring possible.

Alert type	Typical mag	Feasibility / comment
EP-FXT / EP-WXT soft X-ray transient	18 – 20	Automated Gaia-DR3 crossmatch pipeline in place (<code>grb_crossmatch.py</code>).
TDE / SN Ia pre-peak	18 – 20	Routine operation; reference photometry via ZTF DR7.
GW BNS skymap tiling (> 100 deg ²)	—	Not supported at scale (single-FoV telescope, no wide tiling).

6. Data products delivered to collaborators

For every ToO, within the 15-min reaction window and continuously refreshed through the night:

- Calibrated stacked image (`res.fit`, 2×2 binned, WCS-solved)
- STDWeb photometric catalogue (Gaia-BP/RP zero-point, $S/N \geq 5$)
- Transient candidate list (difference imaging against ZTF DR7)
- Light curve (per filter, per visit) — delivered as CSV + interactive HTML
- Provenance: Siril CLI script (`auto_cli.ssf`) reproducing the reduction bit-for-bit

7. How to trigger a ToO

```
POST https://<night-manager>/api/targets/too
Authorization: Bearer <token>
```

```
{
  "name": "AT 2026xyz",
  "ra": 170.9607,
  "dec": 63.6128,
  "priority": "urgent",
  "filters": ["G", "RP"],
  "exp_time": 120,
  "n_frames": 10,
  "broker": "GCN",
  "trigger_id": "EP-FXT-01709259488"
}
```

Accepted payloads return a queue position and ETA; completion triggers a webhook/email to the submitter with the STDWeb result URL. Tokens are issued per collaboration on request.

Observatoire de Bretagne Sud · rapid-response optical follow-up · data calibrated with STDWeb · pipeline source: `astrobatch` (MIT). This note is a *living document* — performance numbers are updated monthly from the rolling STDWeb measurement log.